

UNIT – 1
ASSESSMENT – 2
PROBLEMS

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1. Title

Backtracking Search for Doctor Shift Scheduling

Aim

To assign three doctors (**D1, D2, D3**) to three hospital shifts (**Morning, Afternoon, Night**) using the **Backtracking Search Algorithm** while satisfying all given constraints.

Problem Statement

A hospital needs to assign **3 doctors (D1, D2, D3)** to **3 shifts (Morning, Afternoon, Night)** such that the following constraints are satisfied:

Constraints

1. D1 cannot work the **Night** shift.
2. D2 must work before D3 (**Morning < Afternoon < Night**).
3. Only one doctor can be assigned to each shift.
4. D3 cannot work the **Morning** shift.
5. Every doctor must be assigned **exactly one shift**.

Apply the **Backtracking Search Algorithm** to find a valid assignment, showing all intermediate steps, constraint checks, and the final valid schedule.

2. Title

Robot Path Planning using Breadth-First Search (BFS)

Aim

To find the optimal path for a robot moving in a **5×5 grid** from the Start (S) position to the Goal (G) using the **Breadth-First Search (BFS)** algorithm while avoiding obstacles.

Problem Statement

A robot operates in a **5×5 grid environment**. It must move from **Start (S)** to **Goal (G)**.

Constraints

1. The robot can move **Up, Down, Left, and Right**.
2. Cost of each move = **1**.
3. The robot **cannot pass through obstacles**.
4. Use the **Manhattan Distance heuristic**.
5. Using the above constraints and the **BFS Search Algorithm**, show:
 - Step-by-step node expansion.
 - Queue updates.
 - Optimal path.
 - Total path cost.

3. Title

Autonomous Rescue Robot using Online Search / Uniform Cost Search

Aim

To analyze the decision-making process of an autonomous rescue robot operating in a disaster-affected building and develop an AI-based solution using an appropriate search strategy.

Problem Statement

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a **partially observable environment** where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position **S** and must reach a survivor located at **G**.

Constraints

- The robot can move **Up, Down, Left, Right**.
- Some cells are blocked (obstacles).
- The robot can only observe adjacent cells.
- Each movement has a cost of **1**.
- Entering risky zones adds an additional cost of **2**.
- The robot must minimize the total cost.
- The robot must update its knowledge dynamically (Online Search).

Tasks

- a) Analyze the constraints and explain how they affect the robot's decision-making.
- b) Develop a solution using an appropriate search strategy.
- c) Explain how the robot applies:
- Intelligent Agents
 - Rationality
 - Environment Type
- d) Justify why the chosen approach is the most suitable.